

Variations

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Independent

Abstract

A variation is a theme altered by one constraint. The theme of this body is $1_{(0,1)}$: the indicator of the unit interval. Every paper is a variation on this theme — a different way of looking at the distance d_N^2 , a different angle on what it means to know $1_{(0,1)}$. We catalog the principal variations, identify what each changes and what each preserves, and ask: is there a variation that resolves? A variation that does not preserve the wall but passes through it.

1 The Theme

The theme is:

$$1_{(0,1)} \in L^2(0, \infty).$$

Simple. Complete. Known. The question is how to know it from the integers alone: $\rho_{1/2}, \rho_{1/3}, \rho_{1/4}, \dots$

The BN criterion says: the theme is recoverable from these variations if and only if RH.

Definition 1 (A variation). A *variation* on the BN theme is a modification of one of:

1. the target function (replacing $1_{(0,1)}$ with something nearby),
2. the basis functions (replacing $\rho_{1/k}$ with something related),
3. the inner product (replacing $L^2(0, 1)$ with a weighted space),
4. the index set (replacing $\{2, 3, \dots\}$ with a sparse or dense sequence).

Each variation preserves some structure and changes others. The question is which structure is essential to RH.

2 Variation I: Smoothed Target

Replace $1_{(0,1)}$ with $f_\varepsilon = 1_{(0,1)} * \phi_\varepsilon$, a smooth approximation with $\|f_\varepsilon - 1_{(0,1)}\|_{L^2} < \varepsilon$.

Proposition 2 (Smooth variation converges unconditionally). *For any $\varepsilon > 0$, the smooth BN distance $d_N^\varepsilon = \inf_{f \in \mathcal{V}_N} \|f_\varepsilon - f\|$ satisfies $d_N^\varepsilon \rightarrow 0$ as $N \rightarrow \infty$, unconditionally. The smoother the target, the faster the convergence.*

Proof. The Fourier transform $\hat{f}_\varepsilon(1/2 + it) = \hat{1}_{(0,1)}(1/2 + it)\hat{\phi}_\varepsilon(t)$. With $\hat{\phi}_\varepsilon(t) = e^{-\varepsilon|t|}$, the modified spectral density is $\sigma_\varepsilon(t) = e^{-2\varepsilon|t|}\sigma(t)$, which is integrable on $\mathbb{R}$. By the Szegő–Widom theorem ([2]), $\int \log \sigma_\varepsilon = \int \log \sigma - 2\varepsilon \int |t| = -\infty$, so the continuum prediction error for f_ε is still 0. But the exponential weight suppresses high-frequency zeros, removing the wall: even under $\neg\text{RH}$, $d_N^\varepsilon \rightarrow 0$. $\square$

Remark 3 (Why smoothing removes the wall). The wall comes from zero contributions at all heights γ_k . By smoothing, we exponentially suppress contributions from zeros at height $\gamma > 1/\varepsilon$. There are only finitely many zeros below height $1/\varepsilon$ (unconditionally). These finitely many zeros can be dealt with by a finite number of BN steps. The wall is gone: $d_N^\varepsilon \rightarrow 0$ unconditionally. But the price: we have changed the question. We are no longer asking about $1_{(0,1)}$; we are asking about f_ε . The two differ by ε in L^2 , but ε cannot be sent to 0 unconditionally: as $\varepsilon \rightarrow 0$, the wall returns. The smooth variation does not prove RH; it dissolves the problem.

3 Variation II: Sparse Primes

Replace the index set $\{2, 3, 4, \dots\}$ with the prime set $\{p_1, p_2, p_3, \dots\}$.

Proposition 4 (Prime-only BN flow). *Let $\mathcal{V}_N^P = \text{span}\{\rho_{1/p} : p \leq p_N\}$ be the span of BN functions at primes only. Then $\mathcal{V}_\infty^P$ is dense in $L^2(0, 1)$ (up to the constraint) if and only if RH.*

Proof. The composites are redundant: each $\rho_{1/n}$ with n composite lies in $\overline{\mathcal{V}_\infty^P}$ ([3]). The Gram-Schmidt process applied to composites yields only small new information. Therefore $\overline{\mathcal{V}_\infty^P} = \overline{\mathcal{V}_\infty}$, and density of $\mathcal{V}_\infty$ is equivalent to density of $\mathcal{V}_\infty^P$. Both are equivalent to RH by the Nyman–Beurling criterion. $\square$

Remark 5 (The economy of primes). The prime-only variation shows: the composites are beautiful but not necessary. The ∞ -dimensional span is generated by the primes alone. The composites contribute refinements — correlations, higher-order arithmetic — but not new directions. The BN flow can be run on primes only and reach the same limit. What changes is the rate: the primes alone converge more slowly (each prime step contributes $\cos^2 \theta \sim (\log p)^2/p$; composites contribute smaller but positive corrections). The wall does not change.

4 Variation III: Weighted Inner Product

Replace $\langle f, g \rangle_{L^2(0,1)}$ with $\langle f, g \rangle_w = \int_0^1 f(x)g(x)w(x)dx$ for a weight $w : (0, 1) \rightarrow \mathbb{R}_{>0}$.

Proposition 6 (Weighted BN criterion). *For any continuous weight w bounded above and below, the weighted BN criterion holds: $d_N^w = \inf_{f \in \mathcal{V}_N} \|1_{(0,1)} - f\|_w \rightarrow 0$ if and only if RH. The wall is weight-independent.*

Proof. Since $c_1 \|\cdot\|_{L^2} \leq \|\cdot\|_w \leq c_2 \|\cdot\|_{L^2}$ for a bounded weight, density is preserved. The criterion remains: $\mathcal{V}_\infty$ dense in $L^2(0, 1)$ iff RH. $\square$

Remark 7 (The singular weight variation). The interesting case is a singular weight $w(x) = x^{2\sigma-1}$ that emphasizes the behavior of $x^{-\sigma}$, the critical line modulus. Under RH: $w = x^0 = 1$ (the standard case). Varying σ slightly (taking $\sigma = 1/2 + \varepsilon$ or $\sigma = 1/2 - \varepsilon$) shifts the critical line and changes the problem. For $\sigma > 1/2$: the criterion $d_N^\sigma \rightarrow 0$ is equivalent to ζ having no zeros in $\text{Re}(s) > \sigma$, which is unconditionally true for $\sigma > 1$. Taking $\sigma \rightarrow 1/2^+$ approaches RH from above. The wall is the limit: it appears exactly as $\sigma \rightarrow 1/2$.

5 Variation IV: Shifted Target

Replace $1_{(0,1)}$ with $1_{(0,1)}(\cdot/\theta)$ for $\theta \in (0, 1)$, the target shifted into the interval $(0, \theta)$.

Definition 8 (Beurling's original criterion). The original Beurling–Nyman criterion ([1]) is stated for the family $\{1_{(0,\theta)} : 0 < \theta \leq 1\}$: $\text{RH} \iff 1_{(0,1)} \in \overline{\text{span}}\{x \mapsto 1_{(0,\theta)}(x) : 0 < \theta \leq 1\}$. The functions $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$ are linear combinations of the shifted targets, hence the two formulations are equivalent.

Remark 9 (Variations on the theme). The four variations identify the essential ingredient of the BN criterion:

- The target $1_{(0,1)}$ cannot be smoothed (Variation I).
- The index set can be reduced to primes (Variation II).
- The inner product can be changed (Variation III).
- The target can be shifted but not blurred (Variation IV).

What is essential: the sharp threshold at $x = 1$. The indicator $1_{(0,1)}$ is discontinuous at $x = 1$, and this discontinuity carries all the information about RH. Every variation that preserves this discontinuity preserves the wall. Every variation that blurs it removes the wall — and the question. The theme cannot be varied away.

6 The Unresolved Variation

Theorem 10 (The variation that would prove RH). *The following variation would, if constructed, prove RH: a sequence $f_1, f_2, \dots$ in $\mathcal{V}_\infty$ with $\|1_{(0,1)} - f_N\|_{L^2} \rightarrow 0$, constructed without assuming RH. Such a sequence would establish $d_\infty^2 = 0$ directly.*

No such sequence is known. Every explicit construction either:

1. *converges to f_∞^* (the BN limit, equal to $1_{(0,1)}$ iff RH),*
2. *or requires a hypothesis on the zeros.*

The wall is not a failure of creativity in the variations. It is the structure of the problem: any explicit f_N that converges to $1_{(0,1)}$ must know something about the zeros of ζ , and that knowledge is RH.

References

- [1] Rincón, D. (2026). The Approximation. Preprint.
- [2] Rincón, D. (2026). Widom. Preprint.
- [3] Rincón, D. (2026). Prime. Preprint.
- [4] Rincón, D. (2026). Wisdom. Preprint.
- [5] Rincón, D. (2026). The Rate. Preprint.